

FUNDAMENTALS OF PROMPT ENGINEERING



WHAT IS A PROMPT?

A prompt is the **input** or **instruction** given to an AI model to generate a response.



Renewable energy is clean, sustainable and helps reduce pollution and combat climate change...



KEY TAKEAWAY

The better the prompt, the better the output.



COMPONENTS OF A PROMPT



1. CONTEXT

Provides background information or sets the situation.



6. OUTPUT FORMAT

Specifies how the response should be structured.



5. CONSTRAINTS

Rules, limits, or guidelines for the response.



4. INPUT DATA

Any relevant data, text, or examples that help the AI understand better.



GOOD PROMPTS = **CLEARER INSTRUCTIONS** + **BETTER CONTEXT** + **BETTER RESULTS**

PROMPT DESIGN PRINCIPLES

Well-designed prompts lead to clearer responses, greater accuracy, and more useful outputs.



**KEY TAKEAWAY**
A good prompt is intentional, specific, and user-focused.

GOOD PROMPTS LEAD TO:

Accurate Responses

Time Saving

Better Results

Enhanced Productivity



PROMPT TEMPLATES

Use proven prompt structures to get better, more consistent responses from AI.

TEMPLATE	STRUCTURE	BEST USED FOR	EXAMPLE PROMPT
 1. ROLE + TASK Assign a role and specify the task.	You are a [role]. Your task is to [task].	✓ Simple tasks, summaries, explanations	“ You are a financial advisor. Your task is to explain the benefits of investing in mutual funds. ”
 2. CONTEXT + TASK Provide context and define the task.	Context: [context information] Task: [what you want the AI to do]	✓ When background information is important	“ Context: I am preparing for a product launch. Task: Write a blog post about the importance of customer feedback. ”
 3. INSTRUCTION + FORMAT Give clear instructions and specify the output format.	Instruction: [what to do] Format: [how you want the output]	✓ Structured outputs, reports, tables, code	“ Instruction: List the top 5 data science careers. Format: Provide the information in a table with columns: Career, Description, and Salary. ”
 4. EXAMPLES + TASK Show examples before asking the task.	Examples: [examples] Task: [your task]	✓ Complex tasks, style matching, few-shot learning	“ Examples: Q: 2 + 2 = ? A: 4 Task: Q: 7 + 6 = ? ”
 5. GOAL + CONSTRAINTS + TASK Define goal, constraints and the task.	Goal: [what you want to achieve] Constraints: [limits or rules] Task: [what to do]	✓ High-quality, accurate, constrained responses	“ Goal: Write a blog introduction about AI. Constraints: Max 100 words, no jargon. Task: Write the first paragraph. ”

KEY TAKEAWAY

Use the right template for the right situation and customize as needed.

TIPS FOR BETTER PROMPTS

Be clear and specific

Provide enough context

Set constraints when needed

Iterate and refine



ROLE PROMPTING WITH EXAMPLES

Define a role for the AI to guide its perspective, tone, and depth of the response.



WHAT IS ROLE PROMPTING?

Role prompting means assigning a specific role or persona to the AI before asking your question.

“You are a [role]. Your task is to [task].
Answer in [tone/format] for [audience].” ”



HOW TO USE

- ✓ Choose the most relevant role for your task.
- ✓ Clearly state the role at the beginning of your prompt.
- ✓ Combine with context, instructions, and format.
- ✓ Adjust the role for different perspectives or audiences.



KEY TAKEAWAY

Assigning the right role helps the AI understand the perspective, tone, and depth you expect, resulting in more relevant and accurate responses.

ROLE	PROMPT EXAMPLE	SAMPLE AI RESPONSE
 FINANCIAL ADVISOR	"You are a financial advisor. Explain the benefits of investing in mutual funds for a beginner." Keep it simple."	 Explains how to manage growth
 TEACHER	"You are a high school teacher. Explain photosynthesis in a way that students can easily understand. Use examples."	 Explains water, CO2, and like plants
 SOFTWARE ENGINEER	"You are a senior software engineer. Review the following code and suggest improvements." [Paste code]	 Provides code to improve and best practices
 MARKETING SPECIALIST	"You are a marketing specialist. Write a catchy tagline for a reusable water bottle brand targeting young adults."	 Generates ideas that are eco-friendly
 CONTENT WRITER	"You are a content writer. Write a blog introduction about the future of Artificial Intelligence in healthcare."	 Provides a high-level overview of future trends

BENEFITS OF ROLE PROMPTING



Better
Relevance



Right
Perspective



Improved
Quality



Saves Time



ZERO-SHOT PROMPTING WITH EXAMPLE

Ask the model to perform a task without providing any examples.



WHAT IS ZERO-SHOT PROMPTING?

Zero-shot prompting is the technique of asking the model to generate a response or perform a task without giving any examples.



HOW IT WORKS

- ✓ You provide a clear instruction or question.
- ✓ The model uses its pre-trained knowledge to understand and generate the response.
- ✓ No input-output examples are given.



WHEN TO USE ZERO-SHOT PROMPTING

- When you want quick answers.
- When the task is simple or straightforward.
- When you want the model to use its general knowledge.

EXAMPLE



PROMPT (ZERO-SHOT)

Explain the importance of water conservation.

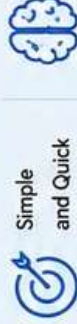


AI RESPONSE (SAMPLE)

Water conservation is essential for sustaining life on Earth. It helps ensure the availability of clean water for drinking, agriculture, and industries. Conserving water reduces wastage, protects natural ecosystems, and helps in combating droughts and water shortages. Small efforts like fixing leaks, using water-efficient appliances, and rainwater harvesting can make a big difference in preserving this precious resource for future generations.



KEY ADVANTAGES



Simple and Quick



Leverages Model's General Knowledge



Saves Time and Effort



Effective for Many Common Tasks



KEY TAKEAWAY

Zero-shot prompting allows the AI to understand and respond to tasks without examples, relying on its pre-trained knowledge.

TIPS FOR BETTER RESULTS



Be clear and specific



Provide enough context



Break complex tasks into simpler steps



FEW-SHOT PROMPTING WITH EXAMPLE

Provide a few input-output examples to guide the model and improve response quality.



WHAT IS FEW-SHOT PROMPTING?

Few-shot prompting means providing a few examples of input and desired output in the prompt to help the model understand the pattern and generate better responses.



HOW IT WORKS

- ✓ You provide several input-output pairs as examples.
- ✓ The model learns the pattern from the examples.
- ✓ It then applies the learned pattern to new inputs and generates the output.



WHEN TO USE FEW-SHOT PROMPTING

- When the task is complex or has specific patterns.
- When you want higher accuracy and consistency.
- When zero-shot prompting does not give satisfactory results.



KEY TAKEAWAY

Few-shot prompting provides examples that act as a guide, enabling the model to understand patterns and produce more accurate and relevant outputs.

EXAMPLES (IN THE PROMPT)

Task: Classify the sentiment of the given review as Positive, Negative or Neutral.
Examples:

Review (Input)		Sentiment (Output)
I really loved the movie! The story was engaging and the acting was excellent.	↑	Positive
The product is of poor quality and broke after two days.	↑	Negative
The food was okay, not great but not bad either.	↑	Neutral
Amazing experience! The staff was very friendly and helpful.	↑	Positive
Shipping took too long and customer service was unresponsive.	↑	Negative



These examples help the model understand the pattern of classifying sentiments.

NEW INFO



Review
The laptop's battery could be better.

MODEL RESPONSE



Sentiment: Neutral

Explanation:

The review is not negative. It mentions the laptop works well, despite a minor aspect (display quality).



Better Accuracy and Consistency



Captures Complex Patterns



Adapts to Specific Requirements



Improves Model Performance



CHAIN OF THOUGHT (CoT) PROMPTING

Encourage the model to think step-by-step to solve complex problems more accurately.



WHAT IS CHAIN OF THOUGHT PROMPTING?

CoT prompting asks the model to break down a problem into intermediate reasoning steps before giving the final answer.



HOW IT WORKS

- ✓ Break the problem into logical steps.
- ✓ Reason through each step sequentially.
- ✓ Combine the steps to reach the final answer.



WHEN TO USE CoT PROMPTING

- Solving math or logic problems
- Multi-step reasoning tasks
- Complex decision making
- When accuracy and explainability are important

EXAMPLE

Task: A shop sells a pen for ₹20. If the shopkeeper gives a 10% discount and then a 5% additional discount, what is the final price of the pen?

CHAIN OF THOUGHT PROMPT

Let's solve this step by step.

- Step 1:** The original price of the pen is ₹20.
Price = ₹20
 - Step 2:** First, a 10% discount is given.
10% of ₹20 = $₹20 \times 0.10 = ₹2$
Price after 1st discount = $₹20 - ₹2 = ₹18$
 - Step 3:** Now, a 5% additional discount is given on the new price (₹18).
5% of ₹18 = $₹18 \times 0.05 = ₹0.90$
Additional discount = ₹0.90
 - Step 4:** Subtract the additional discount from the price after the first discount.
Final price = $₹18 - ₹0.90 = ₹17.10$
- ✓ **Final Answer:** The final price of the pen is ₹17.10.

DIRECT PROMPTING

Prompt:

A shop sells a pen for ₹20. If the shopkeeper gives a 10% discount and then a 5% additional discount, what is the final price of the pen?

Model Response:

CHAIN OF THOUGHT PROMPTING

Prompt:

Let's solve this step by step.
A shop sells a pen for ₹20. If the shopkeeper gives a 10% discount and then a 5% additional discount, what is the final price of the pen?

Model Response (with CoT):

Step 1: ... Step 2: ...

Final Answer: ₹17.10.



KEY TAKEAWAY

CoT prompting improves model performance on reasoning-heavy tasks by guiding it to think step-by-step, just like humans do.



Improves Accuracy on Complex Tasks



Enhances Explainability



Reduces Logical Errors

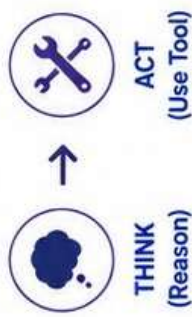


Better Generalization to New Problems



ReAct PROMPTING (Reason + Act)

Combine reasoning and actions (using tools or external information) to solve problems more accurately.



WHAT IS ReAct PROMPTING?



ReAct (Reasoning + Acting) prompting allows the model to reason about a problem, take actions (using tools or APIs), observe the results, and continue reasoning until it can give the final answer.

HOW IT WORKS



- ✓ **Think (Reason):** Understand the question and plan the next step.
- ✓ **Act:** Use a tool or external source to get information.
- ✓ **Observe:** Get the result from the action.
- ✓ **Repeat:** Continue the cycle until the answer is found.

WHEN TO USE ReAct PROMPTING



- Questions that require up-to-date or dynamic information
- Multi-step problems needing external lookups
- Tasks involving calculations, searches, or multiple tools
- When transparency of reasoning and actions is important



KEY TAKEAWAY

ReAct prompting enables LLMs to think, act, observe, and adapt—just like humans do when solving real-world problems.

EXAMPLE: FIND THE CURRENT POPULATION OF JAPAN (2024)

Task: What is the current population of Japan (as of 2024)?



BENEFITS OF ReAct PROMPTING



Improves Accuracy on Complex Tasks



Access to Up-to-date Information



Handles Multi-step Reasoning Effectively



More Transparent and Explainable

ReAct PROMPT TEMPLATE

Use the ReAct format to answer questions that require reasoning and actions.
Use available tools when needed.
Format:
Thought: ...
Action: ...
Observation: ...
... (repeat Thought/Action/Observation)
Final Answer: ...

PROMPT + RESPONSE

Prompt: What is the current population of Japan (as of 2024)?

Response (ReAct Format):

[Shows the same sequence as the example above]

Final Answer:

The current population of Japan is approximately **125.3 million**.



PROMPT PATTERNS

Effective prompt patterns help you get more accurate, relevant and high-quality responses from AI.



1. PERSONA PATTERN

Assign a role or persona to the AI. This helps the model adopt the perspective, knowledge level, tone, and expertise of that persona while generating a response.



HOW TO USE

- ✓ Clearly define the role or persona.
- ✓ Mention the context or domain of expertise.
- ✓ State the task or question.
- ✓ (Optional) Specify the tone, format, or audience.



Prompt:

You are a **senior data analyst** with 10+ years of experience in business intelligence. You are an expert in interpreting dashboards and KPIs.

I will share a sales dashboard summary.

Analyze the key insights, identify trends, and suggest actionable recommendations for improving sales performance.

Explain in a clear and concise manner suitable for business managers.

EXAMPLE: PERSONA PROMPT



AI Response (Example)

Based on the dashboard data, here are the key insights:

- Overall sales increased by 15% in the last quarter.
- The West region contributed 42% of total sales.
- Product Category X showed a 26% growth rate.
- Recommendations:
 - Focus more marketing efforts on Category B.
 - Investigate underperformance in the South region.
 - Upsell complementary products in the West region to increase revenue.



KEY TAKEAWAY

Persona prompting aligns the AI's response with a specific role or expertise, resulting in more context-aware, relevant, and trustworthy answers.





PROMPT PATTERNS

Use the right prompt pattern to get clearer, more accurate and high-quality AI responses.



Re
ref



2. QUESTION REFINEMENT PATTERN

Iteratively improve your question to remove ambiguity, add missing details, and focus the AI on what you really need.



HOW TO USE

- ✓ Start with a basic question.
- ✓ Identify what's missing or unclear.
- ✓ Add context, constraints, or specifics.
- ✓ Refine the question to be more focused.
- ✓ Repeat until the question is clear and well-scoped.

EXAMPLE: REFINED QUESTION



Basic Question (Vague):

"How can I improve employee productivity?"



Refined Question (Better):

"What are the most effective strategies to improve remote employee productivity in a software development team?"



Refined Question (Best):

"What are the most effective, research-backed strategies to improve remote employee productivity in a software development team of 50 people, considering tools, communication, and performance tracking? Provide actionable steps with examples."



AI RESPONSE (to the Best)

Here are effective, research-backed strategies to improve remote employee productivity for a development team of 50 people:

1. **Clear Communication:** Establish regular stand-ups and clear communication channels.
2. **Goal Setting:** Implement clear, measurable goals for the team.
3. **Right Tools:** Use Jira, Slack, and other collaboration tools effectively.
4. **Focus Time:** Encourage dedicated work blocks to minimize distractions.
5. **Recognition & Feedback:** Regularly acknowledge achievements and provide constructive feedback.

Example:

Use a weekly sprint review to identify blockers and celebrate wins.



KEY TAKEAWAY

Refining your question helps the AI understand your true intent and provides more accurate, relevant and actionable answers.





PROMPT PATTERNS

Use the right prompt pattern to get more accurate, reliable and trustworthy AI responses.



Verify
Thoroughly



3. COGNITIVE VERIFIER PATTERN

Ask the AI to verify its own answer by checking assumptions, validating logic, considering alternatives, and confirming accuracy before giving the final response.



HOW TO USE

- ✓ Ask the model to solve the problem.
- ✓ Then ask it to verify the answer.
- ✓ Request it to check assumptions, logic, and calculations.
- ✓ Ask it to consider alternative approaches or edge cases.
- ✓ Finally, confirm and present the best answer with confidence.

EXAMPLE: COGNITIVE VERIFIER PROMPT



Task:

Solve and verify the following:

"What is the total cost of 25 laptops at \$850 each after a 12% discount and 8% tax?"



Prompt:

1. Solve the problem step by step.
2. Verify your answer by checking:
 - Calculations
 - Assumptions
 - Possible alternative interpretations
 - Any potential errors
3. Confirm the final answer with confidence level. If unsure, explain why.

AI RESPONSE



Step-by-Step Solution:

1. Original cost: $25 \times \$850 = \$21,250$
2. After 12% discount: $\$21,250 \times 0.88 = \$18,700$
3. After 8% tax: $\$18,700 \times 1.08 = \$20,196$

Verification:

- Calculations checked: $25 \times \$850 = \$21,250$
- Assumptions: Standard laptop price of \$850
- Alternatives: If tax applied before discount, result would differ (\$19,888)
- Edge cases: No additional discounts or taxes

Final Answer:

\$20,196



Confidence Level: High
The AI correctly calculated the total cost after discount and tax.



KEY TAKEAWAY



The Cognitive Verifier pattern improves accuracy and reliability by making the AI double-check its own work and reduce errors.



PROMPT PATTERNS

Use the right prompt pattern to get more accurate, relevant and high-quality AI responses.



4. TEMPLATE PATTERN

Provide a predefined structure or format for the response.

This helps the AI deliver consistent, well-organized and easy-to-use outputs.



HOW TO USE

- ✓ Define the task clearly.
- ✓ Provide a response template or structure.
- ✓ Specify what each section should contain.
- ✓ (Optional) Add constraints like word limit, tone, or audience.
- ✓ Ask the AI to follow the template strictly.



KEY TAKEAWAY

The Template Pattern ensures structured, consistent and easy-to-use responses—perfect for reports, outlines, summaries and more.



EXAMPLE: TEMPLATE PROMPT



Task:

Create a blog outline on the topic:

"Benefits of Remote Work for Employees and Employers"



Prompt:

Provide the outline using the following template.

Fill each section with key points.

Title: <Blog Title>

Introduction: <Brief overview>

Benefits for Employees:

- Point 1
- Point 2
- Point 3

Benefits for Employers:

- Point 1
- Point 2
- Point 3

Conclusion: <Summary and final thoughts>

Additional Instructions: Keep it concise and easy to read.

AI RESPONSE (EXAMPLE)



Title: The Benefits of Remote Work for Employees and Employers
Introduction:

Remote work has transformed the way we work, offering flexibility and efficiency to both employees and employers.

Benefits for Employees:

- Greater flexibility and work-life balance
- No commuting saves time and reduces stress
- More control over work environment

Benefits for Employers:

- Increased productivity and cost savings
- Reduced overhead and real estate costs
- Access to a wider global talent pool

Conclusion:

Remote work creates a win-win situation, improving productivity, satisfaction, and overall well-being for everyone.



PROMPT CHAINING

Break a complex task into a sequence of linked prompts where the output of one prompt becomes the input for the next.



WHAT IS PROMPT CHAINING?

Prompt Chaining is a technique where multiple prompts are executed in a specific order. Each step builds on the previous output to achieve a final, high-quality result.



HOW IT WORKS

- 1 Break the task into logical steps.
- 2 Create a prompt for each step.
- 3 Use the output of one step as input for the next.
- 4 Continue until the final output is achieved.



BENEFITS

- ✓ Better accuracy by handling complexity step-by-step
- ✓ Improved clarity and focus at each step
- ✓ Easier debugging and refinement
- ✓ More reliable and consistent final output



KEY TAKEAWAY

Prompt Chaining helps you tackle complex tasks by building on each step. It leads to clearer reasoning and more accurate, well-structured results.



EXAMPLE: MARKET ANALYSIS REPORT

1

STEP 1 – RESEARCH

Prompt:
List the key market trends in the renewable energy sector in 2024.

Output:

- Key trends:**
- Growth in solar energy adoption
 - Advancements in battery storage
 - Government incentives & policies
 - Increasing investments
 - Focus on sustainability

2

STEP 2 – ANALYZE

Prompt:
Analyze the above trends and identify opportunities and challenges.

Output:

Opportunities:
High demand, tech innovation, policy support

Challenges:
High initial cost, supply chain issues, regulatory variations

3

STEP 3 – PRIORITIZE

Prompt:
From the opportunities, list the top 3 most impactful ones and explain why.

Output:

1. High demand – expanding market potential
2. Tech innovation – improves efficiency & reduces cost
3. Policy support – ensures long-term growth

4

STEP 4 – RECOMMEND

Prompt:
Based on the top opportunities, provide 3 strategic recommendations for a renewable energy startup.

Output:

1. Invest in scalable solar solutions
2. Partner with battery tech innovators
3. Leverage government incentives for rapid growth

5

STEP 5 – FINAL REPORT

Prompt:
Combine the above into a concise market analysis report.

Final Output:

A well-structured market analysis report with trends, insights, opportunities, challenges, and recommendations.

1
2
3

Solve
step
more



MULTI-STEP PROMPTING

Break a complex task into a sequence of clear steps and guide the AI through each step to get accurate, well-structured results.



Di
Ac
st



WHAT IS MULTI-STEP PROMPTING?

A technique where a complex task is broken into a series of smaller, manageable steps. The output of one step becomes the input for the next step.



HOW IT WORKS

- 1 Break the task into logical steps.
- 2 Create a prompt for each step.
- 3 Use the output of one step as input for the next.
- 4 Continue until the final result is achieved.



BENEFITS

- ✓ Improved accuracy and depth
- ✓ Better control over the output
- ✓ Easier debugging and refinement
- ✓ Ideal for complex reasoning and structured tasks



KEY TAKEAWAY

Multi-Step Prompting helps you solve complex problems by guiding the AI through a series of steps, ensuring clarity, accuracy, and well-structured results.



EXAMPLE: PLAN A 3-DAY TRIP TO JAIPUR

1

STEP 1 – UNDERSTAND THE REQUIREMENTS

Prompt:
List the key things to consider while planning a 3-day trip to Jaipur.

Output:

- Travel dates and duration
- Budget
- Places to visit
- Accommodation
- Food preferences
- Local transport

2

STEP 2 – CREATE ITINERARY OUTLINE

Prompt:
Based on the above, create a day-wise itinerary outline for a 3-day trip to Jaipur.

Output:

- Day 1: Amer Fort, City Palace, Hawa Mahal
- Day 2: Jantar Mantar, Nahargarh Fort, Local Markets
- Day 3: Jaigarh Fort, Jal Mahal, Albert Hall

3

STEP 3 – ADD DETAILS FOR EACH DAY

Prompt:
Provide detailed plan for Day 1 including timings, ticket info, best time to visit, and travel tips.

Output:

- Amber Fort (8:00 AM – 10:30 AM, Entry: ₹550)
- City Palace (11:00 AM – 1:00 PM, Entry: ₹700)
- Hawa Mahal (2:30 PM – 3:30 PM, Entry: ₹50)
- Best time: Morning. Carry water. Wear comfortable shoes

4

STEP 4 – SUGGEST ACCOMMODATION & FOOD

Prompt:
Suggest budget-friendly hotels and must-try local foods in Jaipur.

Output:

- Hotels: Zostel Jaipur, Hotel Polovictory Palace
- Foods: Dal Baati Churma, Ghewar, Pyaaz Kachori, Laal Maas, Rabri

5

STEP 5 – FINAL TRIP PLAN

Prompt:
Combine all details into a final 3-day trip plan summary.

Output:
A complete 3-day Jaipur trip plan with itinerary, timings, places, tickets, hotel suggestions, foods to try, and travel tips.

MULTI-STEP PROMPTING vs CHAIN OF THOUGHT PROMPTING

Both break down complex problems, but they work in fundamentally different ways.

MULTI-STEP PROMPTING		VS		CHAIN OF THOUGHT PROMPTING	
DEFINITION	Breaks a complex task into a sequence of discrete steps. Each step has a specific goal and uses the previous step's output as input.	Encourages the model to think step-by-step through the problem logically before reaching the final answer.			
APPROACH	• Task is divided into sub-tasks or stages. • Each step is a separate prompt or instruction. • Output of one step becomes input for the next.	• Model generates a step-by-step reasoning process in a single response. • Emphasizes logical thinking and intermediate steps. • Leads to the final answer at the end.			
FOCUS	Task decomposition and execution. Getting things done in a sequence.	Reasoning and thinking process. Explaining how the answer is reached.			
HOW IT WORKS	Step 1 Prompt → Output 1 → Step 2 Prompt → Output 2 → ... → Final Output Multiple prompts/steps. Controlled workflow.	"Let's think step by step." 1. First, I will ... 2. Next, I will ... 3. Then, I will ... Therefore, the answer is ... One prompt. Model explains internal reasoning.			
BEST FOR	• Tasks requiring external tools/APIs or data at each step. • Workflows, automation, and structured processes. • When you need control, validation, or edits at each step.	• Problems requiring logical reasoning and math. • Conceptual questions and explanations. • When understanding the reasoning is important.			
EXAMPLE	Goal: Plan a 3-day trip to Manali. 1. Prompt 1: Suggest places to visit. 2. Prompt 2: Suggest hotels for 3 days. 3. Prompt 3: Create day-wise itinerary. 4. Prompt 4: Estimate total budget. Final Output: ✓ Places ✓ Hotels ✓ Itinerary ✓ Budget	Goal: Plan a 3-day trip to Manali. Prompt: "Plan a 3-day trip to Manali. Think step by step." Model Response: 1. First, consider the best places to visit in Manali... 2. Next, plan the day-wise itinerary... 3. Then, choose suitable hotels... 4. Finally, estimate the total budget... Therefore, the complete 3-day plan is ...			
KEY DIFFERENCE	Multi-Step Prompting = <u>External</u> step-by-step execution (You guide the steps)	Chain of Thought Prompting = <u>Internal</u> step-by-step reasoning (The model explains its thinking)			

 **IN SHORT:**

Multi-Step Prompting is about doing steps one by one.
Chain of Thought Prompting is about thinking step by step.

You can even combine both for powerful results.



PROMPT LIMITATIONS

Prompts have limits. Understanding them helps you use AI responsibly and get better results.



Great prompts
but AI is not perfect
Always verify

COMMON LIMITATIONS

	1. Inaccurate or Hallucinated Information	The model may generate facts that sound correct but are wrong or made up.	Example: The model states that "The Great Wall of China is visible from space." (This is false.)
	2. Sensitive to Wording	Slight changes in phrasing can lead to different or unwanted outputs.	Example: "Explain AI simply" vs "Explain AI like I'm 5" → Different results.
	3. Limited Context Window	The model can only consider a limited amount of information at a time.	Example: In a long document, earlier details may be ignored when the limit is exceeded.
	4. Bias and Subjectivity	Outputs can reflect biases present in the training data.	Example: The model may give stereotypical views in summaries.
	5. Lack of Real-Time Knowledge	The model may not have the latest information or real-time updates.	Example: Asking about today's stock prices may return outdated information.
	6. Doesn't Truly Understand	The model predicts the next word/token; it doesn't truly understand or reason.	Example: It may confidently answer a logic puzzle incorrectly.
	7. Overconfidence	The model may present answers confidently even when uncertain or incorrect.	Example: It may give a definite answer with high certainty that it is wrong.



PROMPT LIMITATIONS

Prompts have limits. Understanding them helps you use AI responsibly and get better results.



Great prompts help, but AI is not perfect. Always verify important information.

1. INACCURATE OR HALLUCINATED INFORMATION



AI models can generate information that sounds accurate and confident but is completely wrong or made up. This is called **hallucination**.



WHY IT HAPPENS

- AI predicts the next most likely words, not facts.
- It doesn't have real-world experiences.
- It can't verify truth; it just generates plausible text.



IMPACT

- Spreads false or misleading information.
- Can cause wrong decisions or loss of trust.
- Risky in critical domains (health, law, finance, education).



WHAT YOU CAN DO

- Verify facts from trusted sources.
- Ask for citations or references.
- Cross-check important information.
- Use AI as an assistant, not the final authority.



KEY TAKEAWAY

AI can confidently generate wrong information. Verify, cross-check, and think critically.

EXAMPLE: HALLUCINATED INFORMATION



AI RESPONSE (HALLUCINATION)

Prompt:

Who is the first Indian to win a Nobel Prize in Physics?

AI Response:

The first Indian to win a Nobel Prize in Physics was Homi J. Bhabha in 1954 for his work on nuclear physics.



Why this is incorrect:

Homi J. Bhabha was not a Nobel Prize laureate. No Indian has won a Nobel Prize in Physics yet.



TIP: Treat AI outputs as suggestions. Always verify critical facts before using or sharing the results.



ACCURATE RESPONSE

More Accurate Response:

As of now, no Indian has won a Nobel Prize in Physics. However, several Indian-origin scientists have received Nobel Prizes in Physics, including Subrahmanyan Chandrasekhar and Venkatraman Ramakrishnan.



Best Practice:

- Always verify information from multiple sources.
- Check official websites for critical facts.
- Look for multiple credible references.



Verify today, trust tomorrow. Fact-checking makes you AI-smarter.



PROMPT LIMITATIONS

Understanding limitations helps you write better prompts and get more reliable results.



AI is sensitive to how small changes in wording lead to different or unreliable results.

2. SENSITIVE TO WORDING



Slight changes in phrasing, tone, or structure can lead to different, irrelevant, or incomplete responses.

WHY IT HAPPENS



- Models interpret words and structure literally.
- Ambiguity or vague phrasing can confuse the intent.
- Different word choices may activate different patterns in the model.

IMPACT



- Inconsistent or unexpected outputs.
- Extra time spent refining prompts.
- Risk of incorrect or incomplete information.



KEY TAKEAWAY

Be precise and clear in your wording.
Well-phrased prompts = Better, more consistent outputs.

EXAMPLE: SAME INTENT, DIFFERENT WORDING



PROMPT 1

Explain the causes of climate change.

MODEL RESPONSE

Climate change is caused mainly by greenhouse gas emissions such as carbon dioxide, methane, and nitrous oxide from human activities like burning fossil fuels, deforestation, and industrial processes.



Clear and relevant response.



PROMPT 2

What are the reasons behind climate change?

MODEL RESPONSE

There are various reasons for climate change, including natural variations in Earth's climate, volcanic eruptions, solar activity, and human activities.



Partially correct but includes less relevant information.



Tell me how climate change happens.

MODEL RESPONSE

The various reasons for climate change include natural variations in Earth's climate, volcanic eruptions, solar activity, and human activities.



Tip: Specify what you want, how you want it formatted, and any key constraints.



PROMPT LIMITATIONS

Understanding limitations helps you write better prompts and get more reliable results.



AI can only consider a limited amount of information at a time. Very long inputs may result in details to be missed or forgotten.

3. LIMITED CONTEXT WINDOW



The model can only consider a limited amount of information in a single interaction.



WHY IT HAPPENS

- Every AI model has a maximum context length (token limit).
- When the input exceeds this limit, earlier parts may be truncated or ignored.
- The model may lose track of key details in long conversations or documents.



IMPACT

- Important details may be missed.
- Responses may be incomplete or inconsistent.
- Long documents or conversations may not be fully understood.



KEY TAKEAWAY

Break long inputs into smaller chunks and use multi-step prompting to get complete and accurate results.



Tip: Provide information in parts, ask for clarification, and then combine them for the final answer.

EXAMPLE: LONG DOCUMENT SUMMARY



USER PROMPT (VERY LONG INPUT)

Please read the following 25-page report and provide a summary of key findings, risks, and recommendations.



(Report Content)

Page 1:
Page 2:
.....
Page 24:
Page 25:
(Total ~25 pages / ~40,000+ tokens)



IMPROVED APPROACH

I will share the report in 5 parts, and then provide a final summary.

Part 1

→ Part 2

→ Part 3

→ Part 4

→ Part 5



MODEL RESPONSE (BETTER)

Summary (Comprehensive):

- Part 1: Introduction, background, and purpose.
- Part 2: Methodology, data sources, and analysis.
- Part 3: Financial performance and key metrics.
- Part 4: Risk assessment and mitigation strategies.
- Part 5: Recommendations and conclusions.

Final Summary:

A complete summary covering all key findings, risks, and recommendations from the report.



PROMPT LIMITATIONS

Understanding limitations helps you write better prompts and get more reliable results.



AI models learn from humans. They can have biases and be influenced by groups or situations.

4. BIAS AND SENSITIVITY



AI outputs can reflect biases in training data and may be insensitive or inappropriate in certain cultural, social, or ethical contexts.

WHY IT HAPPENS

- Training data may contain historical, social, or cultural biases.
- AI models don't truly understand context like humans.
- Lack of diverse perspectives in data and design.



IMPACT



- Unfair, discriminatory, or stereotypical outputs.
- Offense or harm to individuals or communities.
- Loss of trust, reputation, and inclusivity.

WHAT YOU CAN DO



- Use inclusive and neutral language in prompts.
- Be aware of potential biases in outputs.
- Review, edit, and validate responses critically.
- Encourage diverse data and perspectives.



KEY TAKEAWAY

AI can be powerful, but it's not always neutral. Be mindful, inclusive, and responsible in your prompting.



Better prompts = Better results =

EXAMPLE: SAME PROMPT, POTENTIALLY BIASED OUTPUT



USER PROMPT

Who is best suited to be a CEO of a company and why?



POTENTIALLY BIASED RESPONSE (EXAMPLE)

A CEO is usually a man with strong leadership qualities, decision-making skills, and experience in business or finance. Men are often more decisive and better suited for handling pressure in top roles.



ISSUE

- Reinforces gender stereotypes.
- Excludes diverse perspectives.
- Can be offensive or harmful.



MORE INCLUSIVE RESPONSE

The best CEO is someone with a strong vision, integrity, and the ability to lead. Success depends on skills, experience, gender, ethnicity, or background, and varied perspectives that benefit the organization.



BETTER APPROACH

- Inclusive and unbiased.
- Focuses on skills and qualifications.
- Respects diversity and promotes equity.



TIP: Always review AI outputs for bias or insensitivity. Refine your prompts, provide context, and provide diverse perspectives.



PROMPT LIMITATIONS

Understanding limitations helps you write better prompts and get more reliable results.



AI models have a knowledge cutoff date. They cannot access live events, or the internet.

5. LACK OF REAL-TIME KNOWLEDGE

AI models are trained on data up to a specific cutoff date and do not have access to real-time information or live updates.



WHY IT HAPPENS

- Models are trained on static datasets.
- No live internet access during response generation (for most models).
- Knowledge is limited to the training cutoff date.



IMPACT

- Outdated or incorrect information.
- Missed recent events, announcements, or changes.
- Risk of making wrong decisions based on old data.



WHAT YOU CAN DO

- Verify information with trusted real-time sources.
- Ask for the cutoff date and validate critical facts.
- Use models with web browsing / plugins if available.
- Combine AI with live data tools or APIs.



KEY TAKEAWAY

AI is powerful, but not always up-to-date. Verify, cross-check, and combine with real-time sources.



Better prompts + Real-time verification

EXAMPLE: ASKING ABOUT A RECENT EVENT



USER PROMPT

Who won the Nobel Prize in Physics in 2024?



MODEL RESPONSE (LIMITATION)

As of my last update in 2023, I don't have information about the Nobel Prize in Physics 2024. Please check the official Nobel Prize website for the latest information.



WHY THIS HAPPENS

- The model's knowledge cutoff is 2023.
- It cannot access or retrieve real-time data or recent announcements.



TIP: Always verify time-sensitive or critical information from reliable, real-time sources.



IMPROVED PROMPT

Who won the Nobel Prize in Physics in 2024? If you don't have real-time access, please provide information on how to find the latest information.



IMPROVED RESPONSE

I don't have real-time access to the latest Nobel Prize in Physics information. You can check the official Nobel Prize website: <https://www.nobelprize.org>

You can also check trusted news sources like BBC, Reuters, or your local news outlet.



PROMPT LIMITATIONS

Understanding limitations helps you write better prompts and get more reliable results.



AI doesn't truly understand the world. It predicts the most likely patterns, not real understanding.

6. DOESN'T TRULY UNDERSTAND



AI models generate responses by predicting patterns in data. They don't have awareness, intentions, or real-world understanding.



WHY IT HAPPENS

- LLMs predict the next word/token, not meaning.
- No consciousness, intentions, or real experiences.
- Struggle with nuance, sarcasm, or implied meaning.



IMPACT

- May produce plausible but incorrect answers.
- Can misinterpret context, humor, or subtle cues.
- Risk of confidently wrong or irrelevant outputs.



WHAT YOU CAN DO

- Provide clear context and explicit instructions.
- Break complex tasks into simpler steps.
- Review, validate, and fact-check the output.
- Use with human oversight and domain knowledge.



KEY TAKEAWAY

AI can generate convincing answers, but it does not truly understand.

EXAMPLE: INTERPRETING SARCASTIC OR COMPLEX CONTEXT



USER PROMPT

Friend: "Great! Another Monday morning... just what I needed."
What does my friend mean?



MODEL RESPONSE (LIMITATION)

Your friend is happy about Monday morning and feels positive. They are excited to start the week.



WHY THIS HAPPENS

- The model processes words literally.
- It may not detect sarcasm, tone, or intent without explicit cues.
- Lacks real-world experiences and emotions.



IMPROVED PROMPT

My friend said: "Great! Another Monday morning... just what I needed."
They said this in a tired and sarcastic tone. What do they likely mean?



IMPROVED RESPONSE

Your friend is likely being sarcastic. They probably mean they are tired of Monday morning and feel tired about starting the week.



TIP: Be specific, add context, and clarify intent. AI can assist, but human judgment is essential.



Better prompts + Human validation = More accurate results



PROMPT LIMITATIONS

Understanding limitations helps you write better prompts and get more reliable results.



AI models can be overconfident.
They may present information with strong confidence, but it's not always accurate.

7. OVERCONFIDENCE



AI may present answers in a confident tone, even when the information is uncertain, speculative, or incorrect.



WHY IT HAPPENS

- Models predict the most likely answer.
- Trained to generate fluent and decisive responses.
- No internal awareness of certainty or doubt.



IMPACT

- Users may trust incorrect information.
- Can lead to poor decisions or risky outcomes.
- Difficult to know what the model is unsure about.



WHAT YOU CAN DO

- Ask for confidence level or sources.
- Cross-check important facts.
- Encourage the model to say "I'm not sure" when uncertain.
- Use the model as an assistant, not final authority.



KEY TAKEAWAY

AI can sound confident, but that doesn't mean it's always correct.
Question, verify, and think critically.

EXAMPLE: ASKING AN UNCERTAIN FACT



USER PROMPT

Who will win the 2035 Cricket World Cup?
(As of 2024, the tournament hasn't happened yet.)



OVERCONFIDENT RESPONSE (LIMITATION)

India will win the 2035 Cricket World Cup.
They have a strong team with excellent players and leadership, which gives them the highest chance of winning the tournament.



WHY THIS IS A PROBLEM

- The event hasn't happened yet.
- The model is making a speculative claim as if it is certain.
- No acknowledgement of uncertainty.



TIP: Ask the model to indicate confidence, cite sources, and mention uncertainty.
Always verify critical information from reliable sources.



BETTER RESPONSE (IMPROVEMENT)

The 2035 Cricket World Cup has not happened yet, so it's impossible to know who will win. Many factors like team form, player injuries, and conditions will influence the outcome. I can share top contenders or past winners you'd like.



WHY THIS IS BETTER

- Acknowledges uncertainty
- Provides helpful context.
- Avoids false certainty.



Better prompts + Verification + Critical Thinking
= Smarter and safer use of AI.

